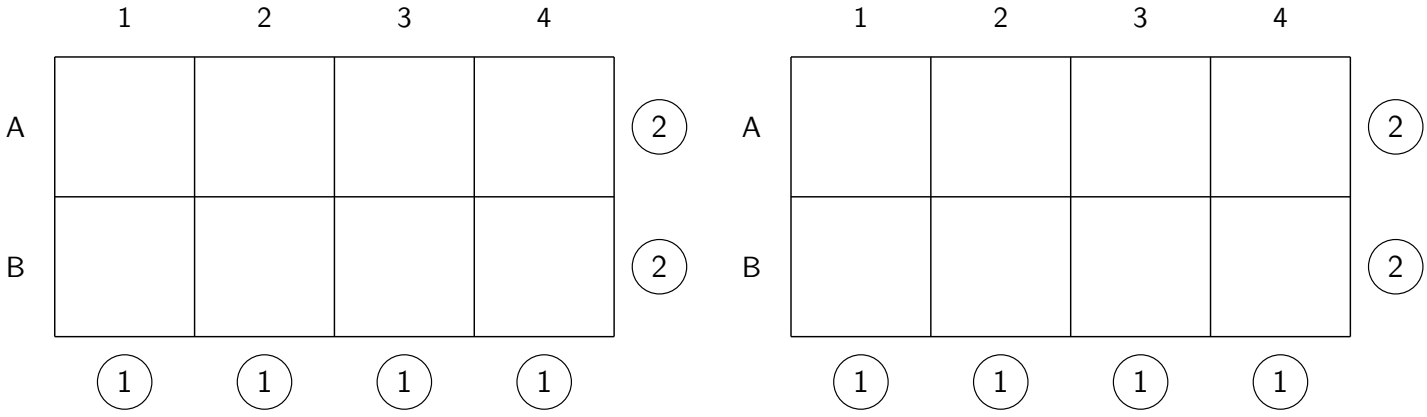
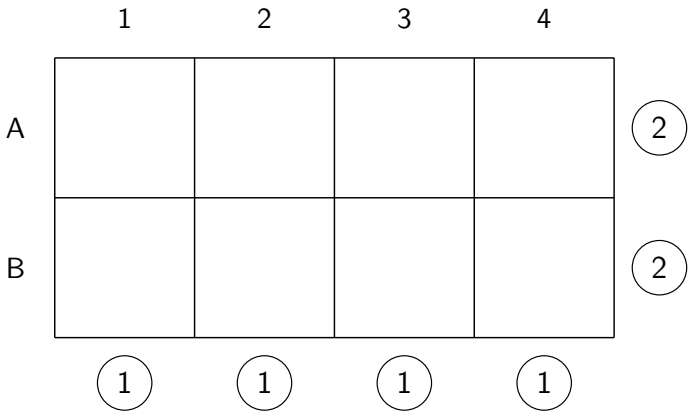
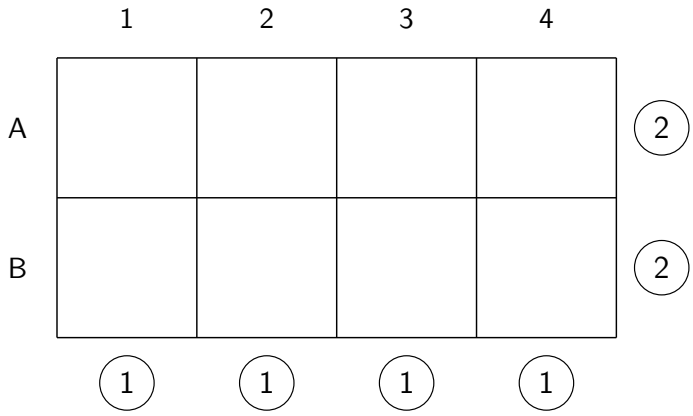
Each circled number counts the counters in its whole row or column. Put at most one counter in a cell. Two pictures are different if any cell is different.

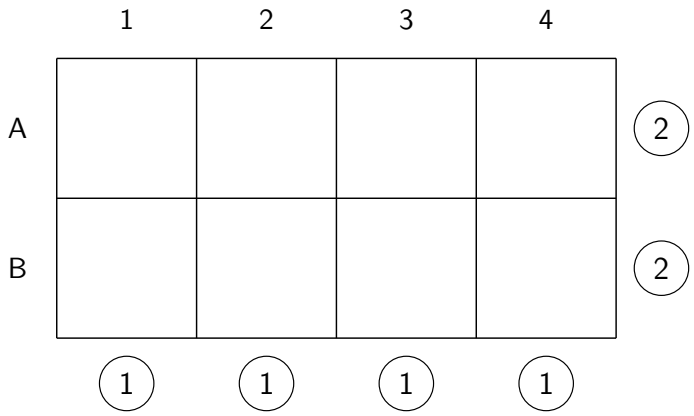
Problem 1: Find and draw every different picture with these counts. Explain why your list is complete.

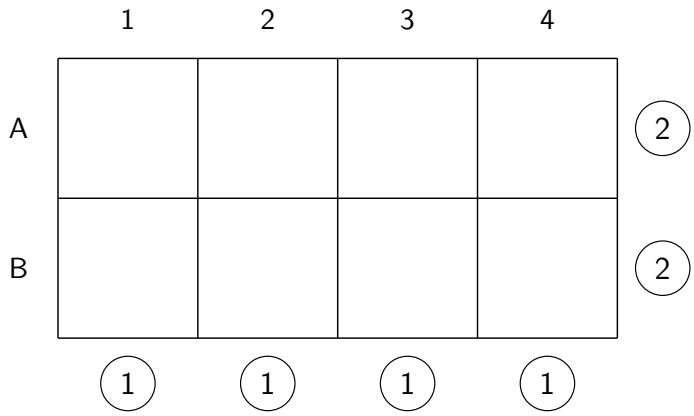


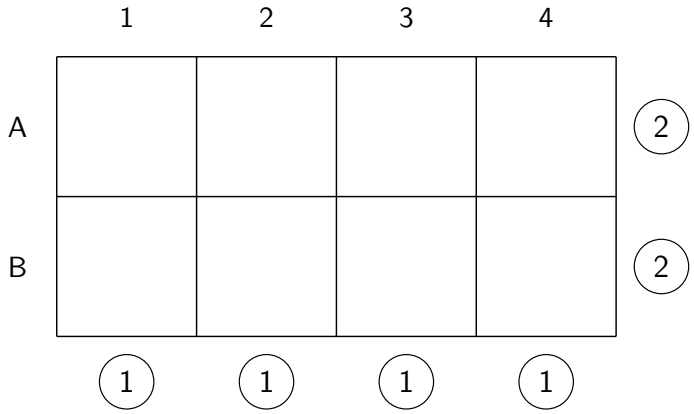


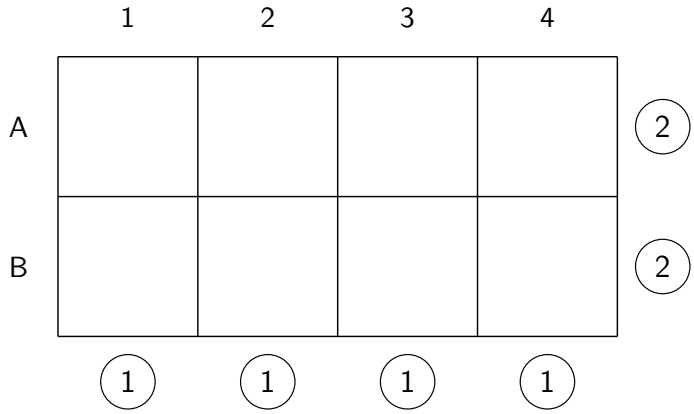


Problem 1 (continued)



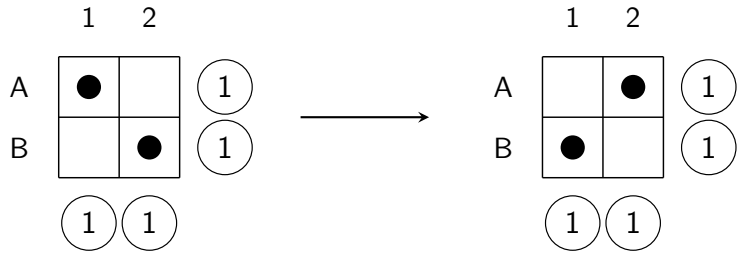


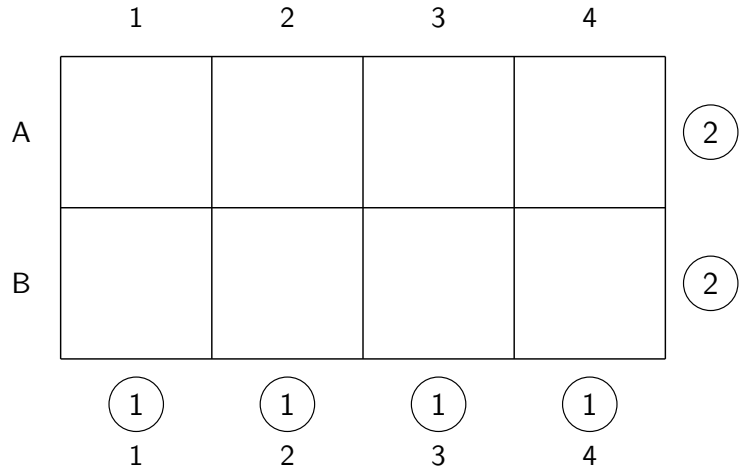


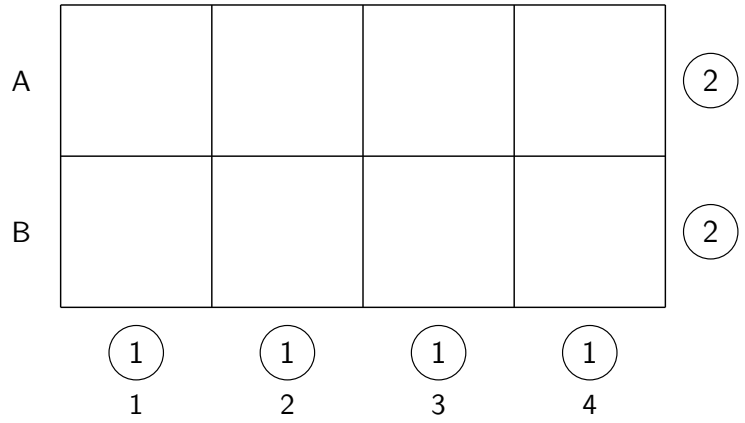


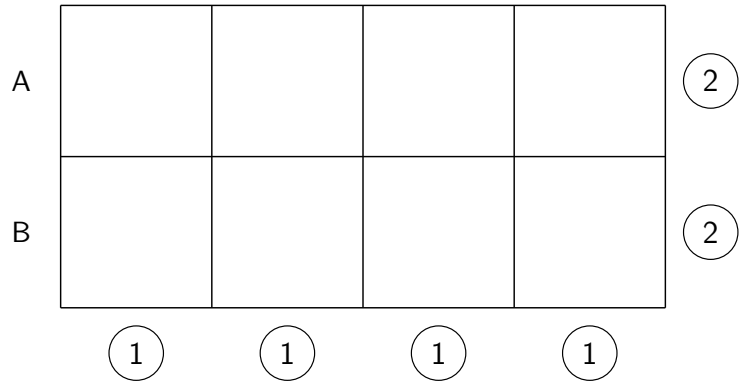
Problem 2: A switch moves the two counters to the empty corners shown; other cells stay fixed. Any two rows and columns may be used. Can any picture from Problem 1 reach any other by switches? What is the largest number of switches ever needed between two of them?

One switch:

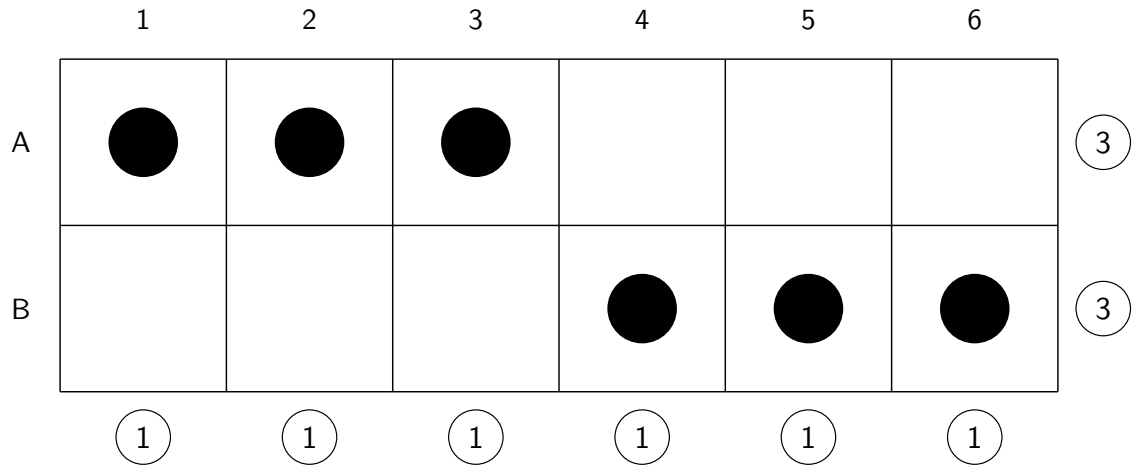


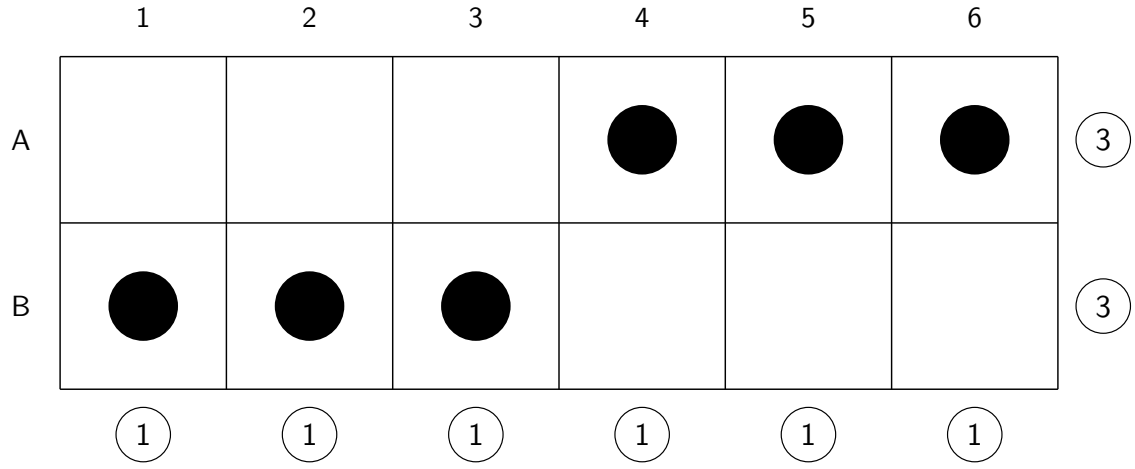




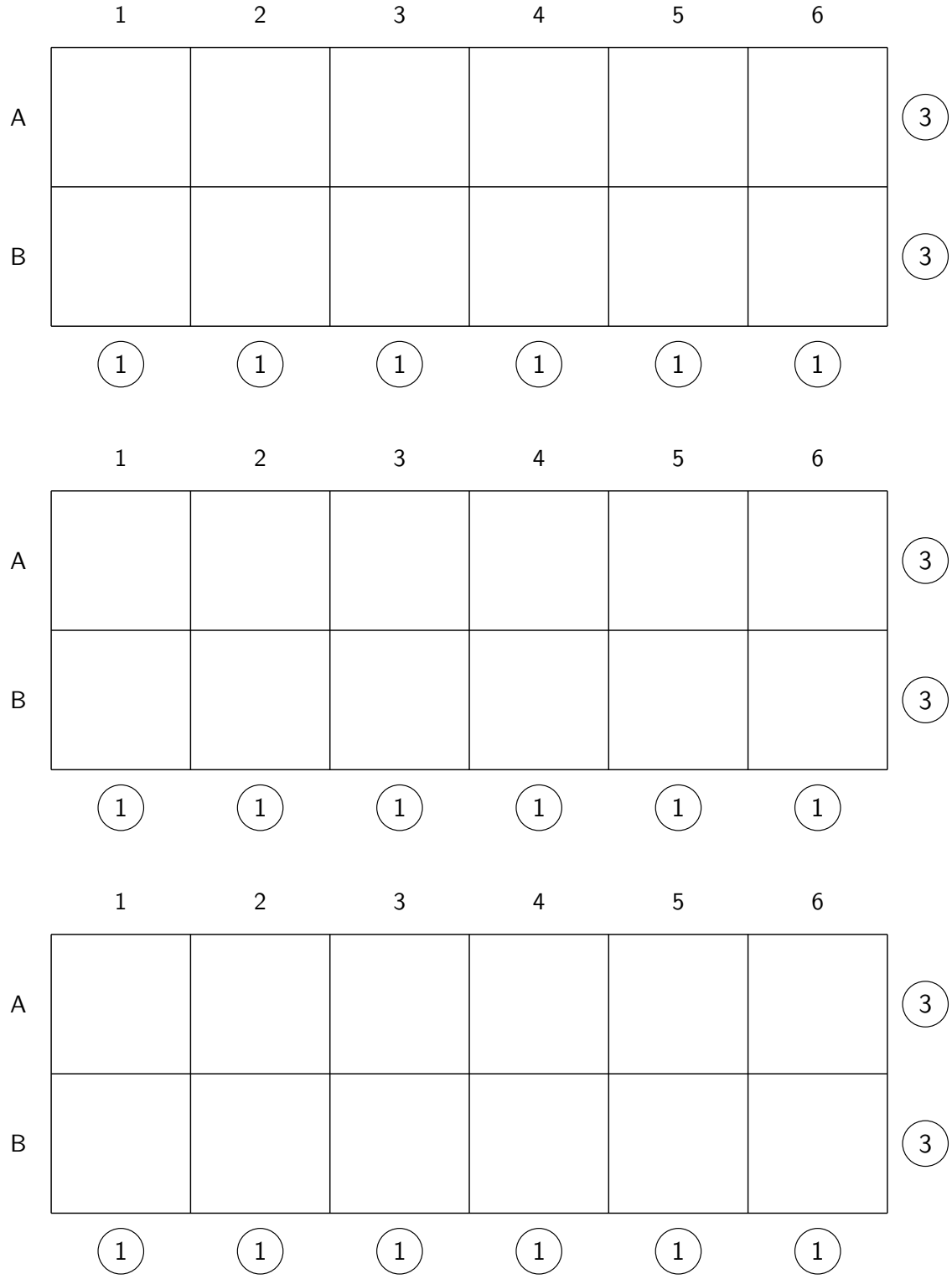


Problem 3: Change the upper picture into the lower picture in as few switches as you can. Record a shortest route and explain why fewer switches cannot work.

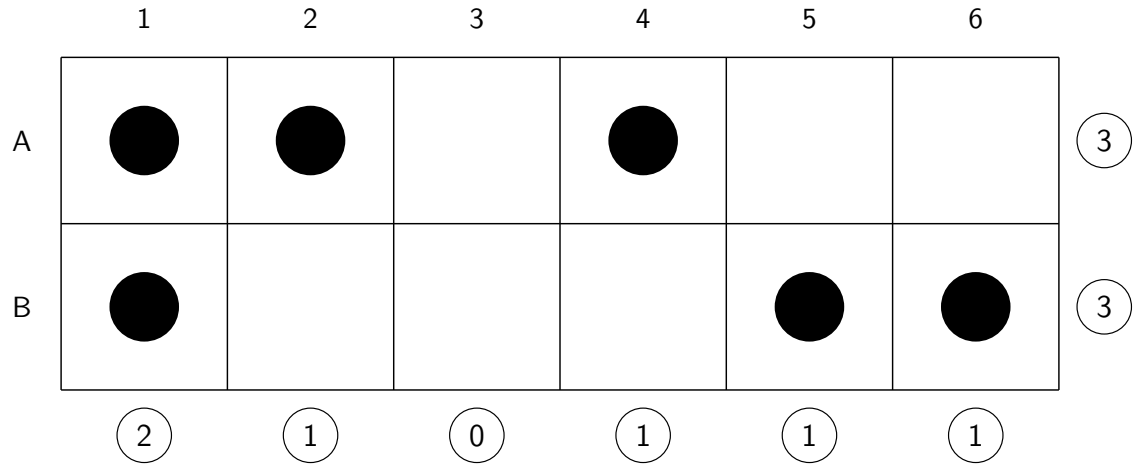


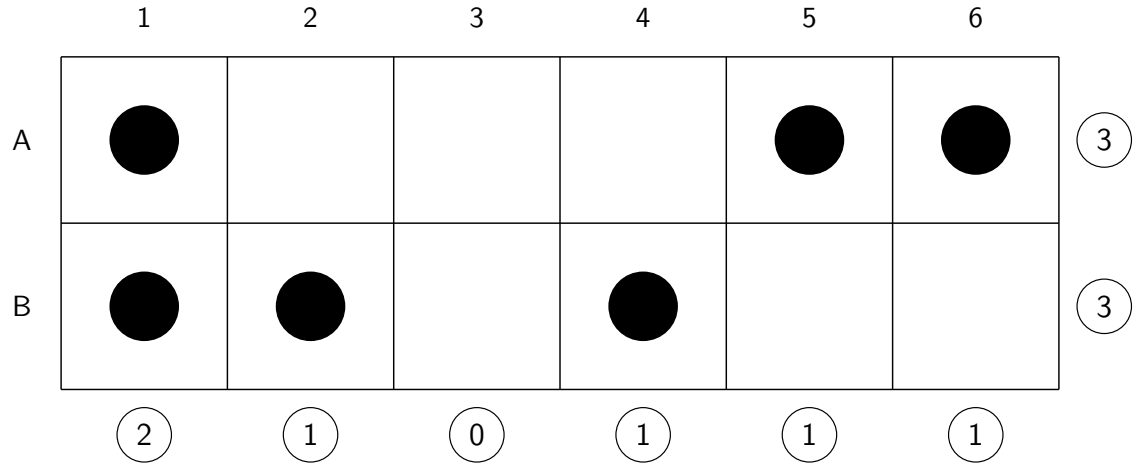


Problem 3 (continued)

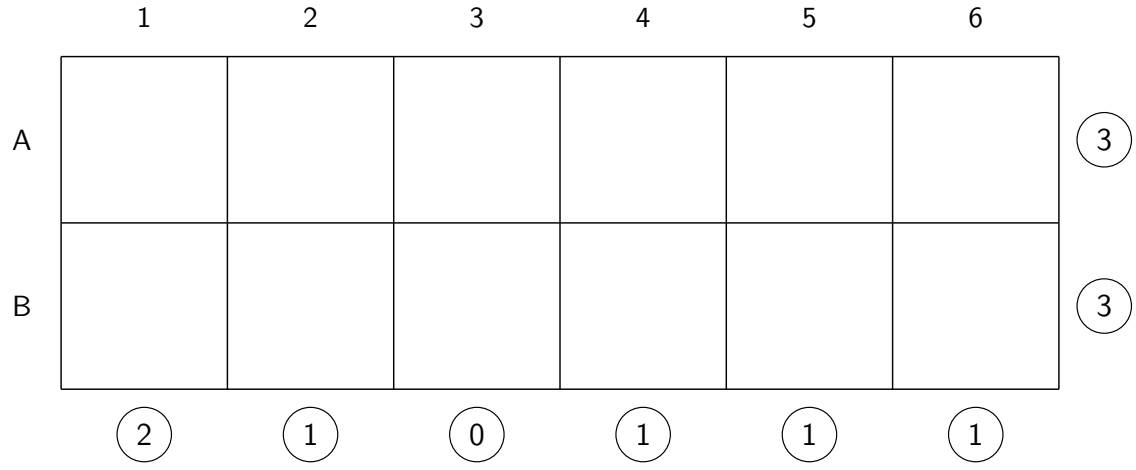


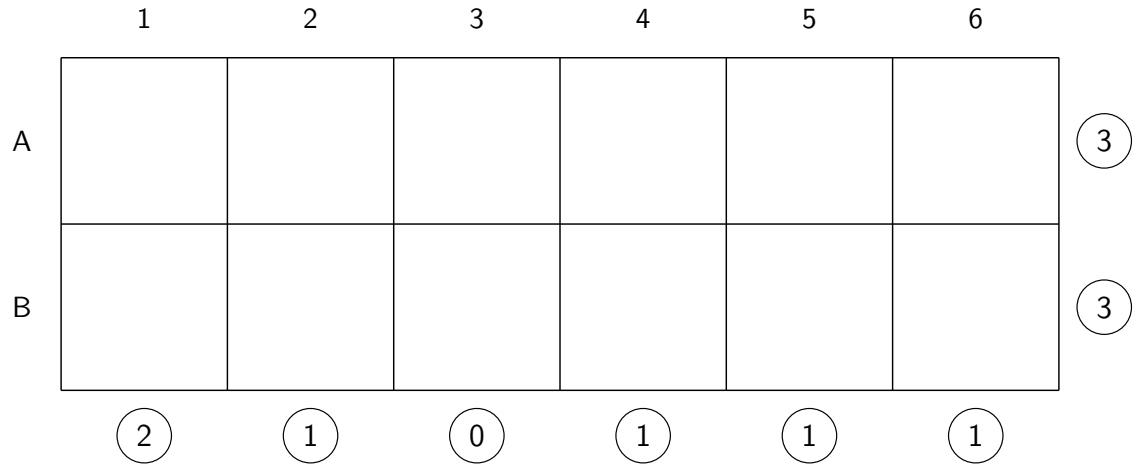
Problem 4: Find the fewest switches from the upper picture to the lower picture. How many different pictures share these counts? What is the largest number of switches ever needed between two of them?

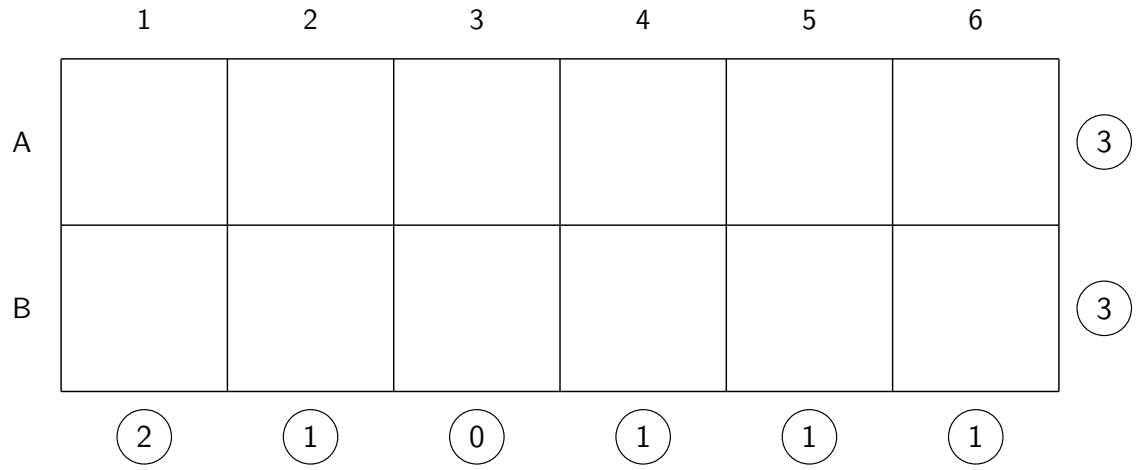




Problem 4 (continued)







Problem 5: Take any two pictures on a two-row grid with the same row and column counts. Explain why switches can always turn one into the other. Give a rule for the fewest switches needed, using only the two pictures. Your explanation should work for any number of columns.

